ELECTIVE COURSE-8.4

UEC825: MEMS				
	L	T	P	Cr
	3	0	0	3.0

Course Objective: To educate the student to understand the fundamentals of Micro Electro Mechanical Systems (MEMS), different materials used for MEMS, semiconductors and solid mechanics to fabricate MEMS devices, various sensors and actuators, applications of MEMS to disciplines beyond Electrical and Mechanical engineering.

Syllabus

Introduction: History of Micro-Electro-Mechanical Systems (MEMS), Market for MEMS, MEMS materials: Silicon, Silicon Dioxide, Silicon Nitride, Polysilicon, Silicon Carbide, Polymers, Thin metal films, Clean rooms.

Process Technologies: Wafer cleaning and surface preparation, Oxidation, Deposition Techniques: Sputter deposition, Evaporation, Spin-on methods and CVD, Lithography: Optical, X-ray and E-Beam, Etching techniques, Epitaxy, Principles of bulk and surface micro machining, Lift-off process, Doping: Diffusion and Ion Plantation, Wafer Bonding: Anodic bonding and Silicon fusion bonding, Multi User MEMS Process (MUMPs), Introduction to MEMS simulation and design tools, Lumped element modeling and design, Electrostatic Actuators, Electromagnetic Actuators, Linear and nonlinear system dynamics.

Sensing and Actuation Principles: Mechanical sensor and actuation: Principle, Beam and Cantilever, Microplates, Capacitive effects, Piezoelectric Materials as sensing and actuating elements, Starin Measurement, Pressure measurement, Thermal sensor and actuation, Micro-Opto- Electro mechanical systems (MOEMS), Radio Frequency (RF) MEMS, Bio-MEMS.

Application case studies: Pressure Sensor, Accelerometer, Gyroscope, Digital Micromirror Devices (DMD), Optical switching, Capacitive Micromachined Ultrasonic Transducers (CMUT)

Course Learning Outcomes (CLOs)

The students will be able to:

1. integrate the knowledge of semiconductors and solid mechanics to fabricate MEMS devices
2. apply different materials used for MEMS
3. design the micro devices using the MEMS fabrication process
4. analyze operation of micro devices, micro systems and their applications

Text Books

1. Introduction to Micro Fabrication, Franssila Sami, WILEY, 2nd Edition, 2010
2. An Introduction to Microelectromechanical Systems Nadim Maluf, Engineering, Artech House, 3rd edition, 2000.
3. MEMS, Mahalik Nitaigour Premchand, McGraw-Hill, 2007.

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 113th meeting of the Senate held on March 11 and September 7, 2024, respectively. Further, revision Approved in 1st meeting of Academic Council held on September 11, 2025.

Reference Books

Microsystem Design, Senturia Stephen D., Springer US, (2013).

1. Fundamentals of Microfabrication, Madou Marc J., CRC Press, (2002).
2. MEMS Mechanical Sensors, StephrnBeeby, Graham Ensell, Michael Kraft, Neil White, artech House (2004).
3. Foundations of MEMS, Chang Liu, Pearson Education Inc., (2012)
4. MEMS& Micro systems Design and Manufacture Tata McGraw Hill, Tai Ran Hsu, NewDelhi, 2002.

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessional (May include Assignments//Quizzes)	25

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